

Impact of Reconstruction Errors on Connectome Graph Analysis

Experimental Analysis of Missed Synapses and False Synapses

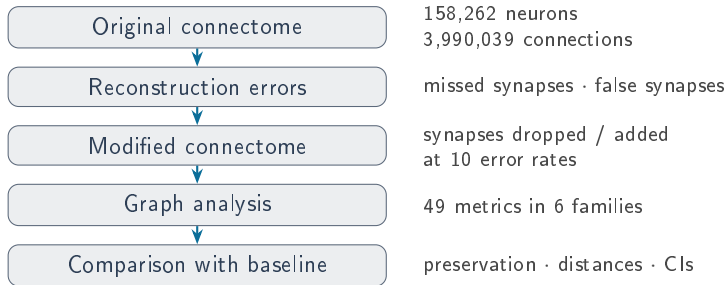
BANC Dataset

Neuron- and synapse-resolved connectome simulation study · 10 error rates · 5 trials each

The Problem

Connectomes are reconstructed from noisy imaging data

Every neuron and every synapse in a connectome is the output of an automated segmentation / synapse-detection pipeline — and pipelines make mistakes.



Question

How much do these errors distort the results of graph analysis — and do different error types distort it differently?

Research Questions

- ▶ **Direct effect.** How do total synapses and connections change as the error rate grows?

Expected: proportional to the rate — but by how much for edges?

- ▶ **Structural effect.** Are global properties (connectivity, reciprocity, hub ranking) preserved, or do they degrade?

Does the coarse wiring plan survive 20% error?

- ▶ **Comparative effect.** Which error type — missed synapses (false negatives) or false synapses (false positives) — is more disruptive at the same nominal rate?

Same rate, different fingerprints?

Deliverable

A data-backed comparison of two error models on the BANC connectome: what degrades first, what survives, and how to tell the models apart.

Dataset and Baseline

Dataset	BANC — <i>D. melanogaster</i> brain + nerve cord
Neurons	158,262
Connections	3,990,039
Synapses	23,556,214
Reciprocity	0.1299
Giant SCC	139,405 vertices
Mean edge weight	5.90 synapses

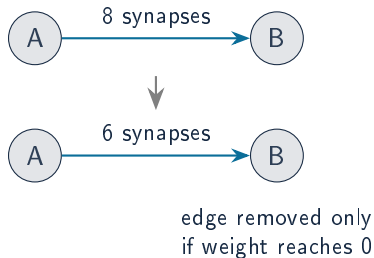
Baseline role

The original graph is the *ground truth*. Every experiment perturbs this graph, re-runs the analyses, and compares the result with the baseline — a controlled, paired design.

Why BANC?

A full adult brain + nerve cord connectome: large enough to be realistic, small enough to run 100 perturbation trials.

Error Model 1: Missed Synapses (False Negatives)



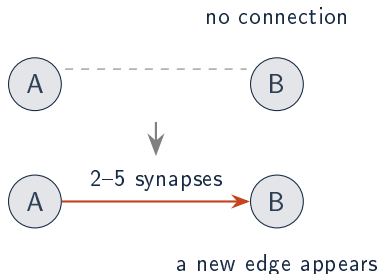
Mechanism

Each synapse survives independently with probability $1 - r$ (binomial sampling). The edge weight becomes the number of surviving synapses.

Consequences

- ▶ Synapse count drops $\approx r$ (linear).
- ▶ Edge count drops *slowly*: an edge survives while at least one synapse remains.
- ▶ Topology (which neurons connect) is largely preserved.
- ▶ r swept over 10 rates: 0 – 20%; seeded, 5 trials per rate.

Error Model 2: False Synapses (False Positives)



Mechanism

$k = r \times (\text{edge count})$ artificial connections are injected between neuron pairs with *no* connection; weights drawn from the weak-edge distribution (≤ 5 synapses).

Consequences

- ▶ Edge count grows $\approx r$ (linear).
- ▶ Injected edges are *weak* (≈ 3 synapses vs. baseline mean 5.9).
- ▶ No new strong hubs are created — but the graph is inflated.
- ▶ Same 10 rates (0 – 20%), same seeds, 5 trials per rate.

Experimental Design

Error rates	0, 0.25, 0.5, 0.75, 1, 2, 5, 10, 15, 20%
Trials	5 seeded per rate
Total runs	2 models $\times$ 10 rates $\times$ 5 trials
Metrics	49 in 6 analysis families
Runtime	$\approx$ 3 min / trial ($\approx$ 5 h total)

Pipeline (fixed order)

Load $\rightarrow$ Build $\rightarrow$ Preprocess $\rightarrow$
Perturb (error model) $\rightarrow$
Analyze $\rightarrow$ Statistics $\rightarrow$ Export

Analysis families

structure $\cdot$ degree distributions $\cdot$ SCC/WCC $\cdot$
reciprocity $\cdot$ assortativity $\cdot$ PageRank

Reproducibility

All stochastic operations use recorded seeds; trials differ only by the seeded sampling.

How is Preservation Measured?

Symmetric preservation ratio

Every graph metric is converted to a preservation score between the baseline value B and the perturbed value P :

$$\text{Preservation (\%)} = \frac{\min(B, P)}{\max(B, P)} \times 100$$

A metric that does not change scores 100%; a 20% loss scores 80%.

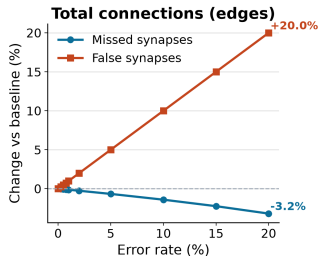
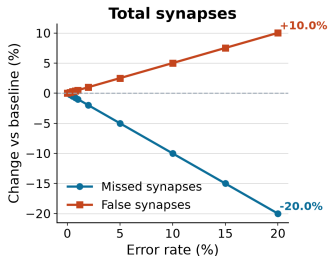
Which metrics qualify?

14 of 49 metrics have a well-defined baseline (node/edge counts, weights, reciprocity, assortativity, components) and enter the preservation analysis. The other 35 (KS / Wasserstein distances, PageRank score rows) have baseline 0 — the symmetric ratio is meaningless — so they are reported as *distances* instead.

Status labels

Preserved $\geq 99\%$ · Minor $\geq 95\%$ · Moderate $\geq 90\%$ · Significant $< 90\%$

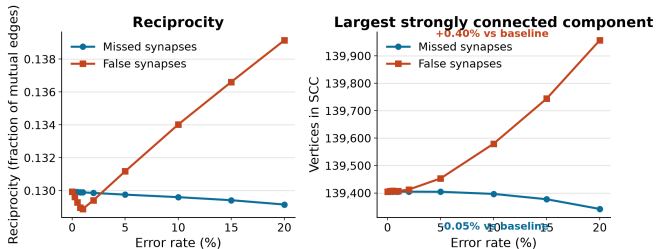
Result 1: Direct Effects are Proportional to the Rate



- ▶ **Missed:** synapses -20.0% at 20% rate — exactly proportional; edges only -3.2% .
- ▶ **False:** edges $+20.0\%$, synapses $+10.0\%$ — again proportional.
- ▶ Both trends are linear across all ten rates.

Takeaway: Both models act exactly as calibrated: the perturbed quantity (synapses or edges) moves in proportion to the error rate — a clean, predictable first-order signal.

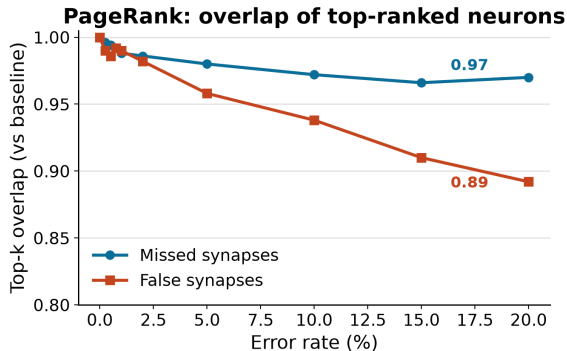
Result 2: The Coarse Wiring is Highly Robust



- ▶ **Giant SCC** keeps 99.95% of vertices under missed; grows 0.40% under false — essentially unchanged.
- ▶ **Reciprocity** moves only -0.6% (missed) / $+7.1\%$ (false).
- ▶ The graph stays connected and its wiring plan survives.

Takeaway: Even at 20% error, the global architecture — connectivity and reciprocity — is barely disturbed by either error type.

Result 3: Hub Ranking Survives, False Synapses Erode It More

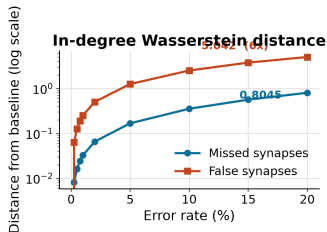
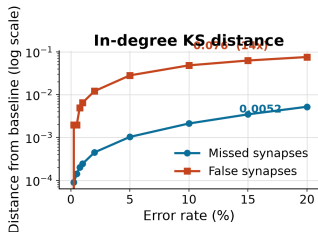


- ▶ **Correlation** with baseline PageRank: 0.9975 (missed) vs. 0.9833 (false) at 20%.
- ▶ **Top- k overlap** of ranked neurons: 0.970 (missed) vs. 0.892 (false) at 20%.
- ▶ The identity of the most important neurons is largely preserved; injected weak edges add ranking noise without creating hubs.

Takeaway: Hub ranking is robust under both models, but false synapses erode it noticeably more (top- k 0.89 vs. 0.97).

Result 4: False Synapses Distort Degree Distributions Far More

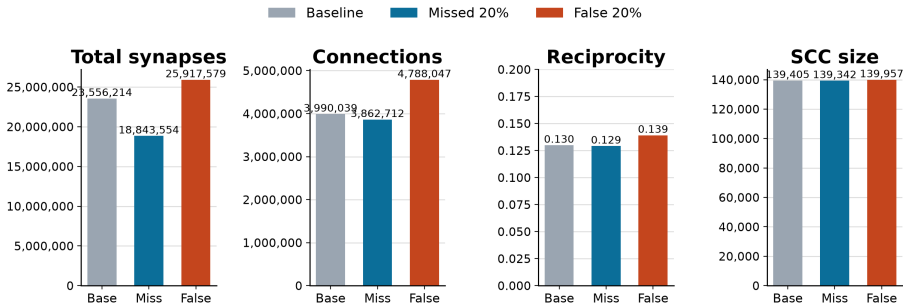
False synapses distort degree distributions 6–14x more at 20%
(KS 14x, Wasserstein 6x)



- ▶ In-degree **KS distance** at 20%: 0.0052 (missed) vs. 0.0758 (false) — **14.5x** larger.
- ▶ **Wasserstein** distance: 0.80 vs. 5.04 — **6.3x** larger.
- ▶ Log scale keeps both curves readable (they span $>10\times$).

Takeaway: At the same nominal rate, false synapses distort the degree distribution an order of magnitude more than missed synapses — the single largest difference between the two models.

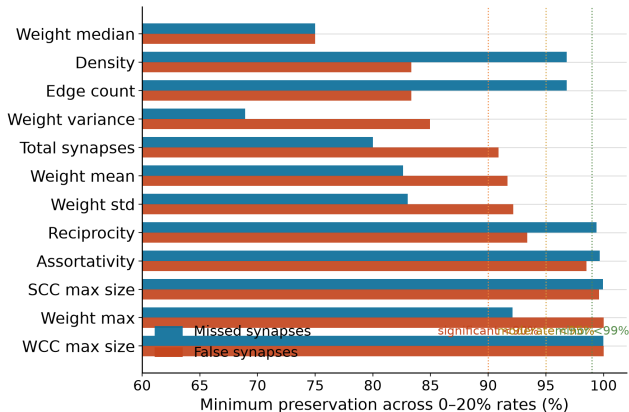
Result 5: Snapshot at 20% Error



Baseline (gray) vs. 20% missed (blue) vs. 20% false (orange). Note the plain (non-scientific) tick labels and honest zero-based scales.

Takeaway: Synapses shrink by a fifth under missed; edges grow by a fifth under false. Reciprocity and SCC size move only a few tenths of a percent.

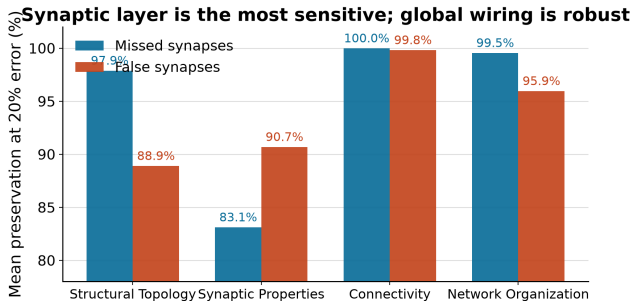
Result 6: What Degrades First? — Sensitivity Ranking



- ▶ **Missed:** weight statistics degrade first — variance 68.9%, median 75.0%, synapses 80.0%.
- ▶ **False:** edge count / density degrade first — 83.3%.
- ▶ Connectivity, reciprocity, assortativity: all $\geq 93\%$.
- ▶ Threshold guides: significant $< 90\%$, moderate $< 95\%$, minor $< 99\%$.

Takeaway: The synaptic layer erodes first under missed synapses; the edge layer erodes first under false synapses. Global wiring is last to move.

Result 7: Preservation by Biological Category at 20%



Category	N	Missed	False
Structural Topology	3	97.9%	88.9%
Synaptic Properties	7	83.1%	90.7%
Connectivity	2	100.0%	99.8%
Network Organization	2	99.5%	95.9%

Unweighted mean of member-metric preservation scores at the 20% rate.

Takeaway: Category-level view confirms the ranking: synaptic statistics and structural topology are the sensitive layers; connectivity is untouched.

Comparison: Missed vs. False at 20% Error

Metric @20%	Missed	False
Total synapses	−20.0%	+10.0%
Edges	−3.2%	+20.0%
Reciprocity	−0.6%	+7.1%
SCC giant	−0.05%	+0.40%
PageRank corr.	0.9975	0.9833
Top- <i>k</i> overlap	0.970	0.892
In-degree KS	0.0052	0.0758

Same nominal rate → different fingerprints

Missed synapses *deflate* the synapse pool while leaving edges and topology intact.

False synapses *inflate* the edge count and distort degree distributions — the more damaging of the two.

Reliability of the Results

- ▶ **5 seeded trials** per rate $\times$ 10 rates $\times$ 2 models.
- ▶ **Tight spread:** total-synapse σ is only $\approx 0.01\%$ of the mean at 20%.
- ▶ **Deterministic false edges:** injected edges reproduce exactly ($\sigma = 0$); missed trials agree to within $\approx 0.01\%$.
- ▶ **Monotone primary trends** (synapses, edges, KS) across all ten rates — no flips.
- ▶ **Consistent status labels** across trials (Preserved $\geq 99\%$ / Minor $\geq 95\%$ / Moderate $\geq 90\%$ / Significant $< 90\%$).

Spread @20% — σ/mean (rel.)

Metric	Missed	False
Total synapses	$\pm 0.009\%$	$\pm 0.005\%$
Edge count	$\pm 0.005\%$	$\pm 0.000\%$
SCC giant	$\pm 0.010\%$	$\pm 0.006\%$

Conclusions

Direct, measurable effects

Missed synapses remove synapses *exactly* in proportion to the rate (-20.0% at 20%); false synapses add edges *exactly* in proportion ($+20.0\%$). These are the most sensitive quantities measured.

Topology is highly robust

Giant component and node count survive to $>99\%$ even at 20% error — the coarse wiring plan is barely disturbed (PageRank correlation ≥ 0.98).

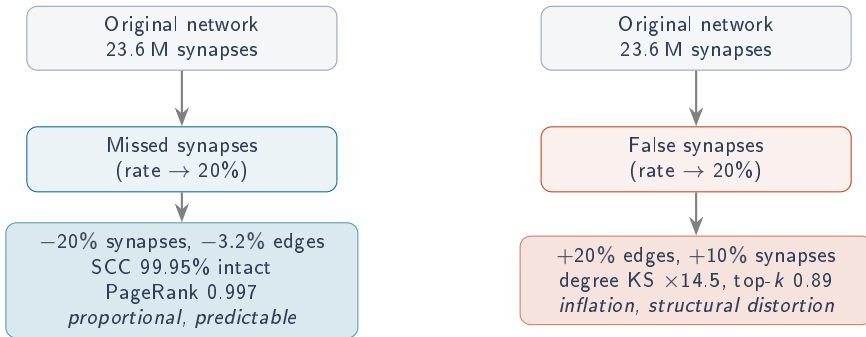
False synapses are the more disruptive model

At equal nominal rates they distort degree distributions $6\text{--}14\times$ more, erode hub ranking (top- k 0.89 vs. 0.97) and raise reciprocity by $+7\%$.

Error signatures are separable

Missed synapses *deflate* synaptic statistics; false synapses *inflate* the graph — weak edges dilute weights without hubs.

Summary



Takeaway: reconstruction errors degrade synaptic statistics first — global wiring stays intact. False-positive errors are consistently more disruptive than missed synapses at the same rate.